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ASSIGNMENT

1. (i) Semantics

Meaning of statements in a programming language

ii) Syntax - Rules defining structure of valid statements in a programming language

iii) Logic - sequence of operators used to solve a problem in code

2. Procedure & function

Procedure → Performs task but does not return a value

```
Sub Display Message()
  Console.WriteLine("How are you,")
End Sub
Module Module1
    Sub main()
      Display Message()
End Sub
End Module
```

function → Performs a task and returns a values

```
function AddNumbers (a As Integer, b As integer) As integer
  Return a+b
End function
Module Module1
  Sub Main()
    Dim result As Integer = AddNumber (5,3)
    Console.WriteLine ("sum : " result)
End Sub
End Module
```

ii. Variable and Array

Variable → A variable store single value

```
Module Module 1
  Sub Main ()
  Dim number as integer = 10
   Console.Writeline ("Value" & number)
 End Sub
 End Module
```

Array ⇒ store multiple value of same data type in a single variable

```
Module Module 1
  Sub Main()
   Dim numbers (2) As integer
   numbers (0) = 1
   number (1) = 2
   number (2) = 3
 for i As integer = 0 to 2
   Console Writeline ("Element " & i & ": " & number (i))
   Next
 End sub
   End Module
```

3. Decision Making Statements and Loop Statements

Decision making execute based on condition e.g if Else

```
Module Module 1
  Sub Main ()
   Dim Age As integer = 20
    If Age >= 18 then
    Console.Writeline ("You are an adult.")
  Else
    Console Writeline ("You are a minor.")
End If
End sub
End Module.
```

Loop statements - repeats code multiple times e.g for while

```
Module Module 1
Sub Main ()
    for i as integer = 1 to 5
        Console.WriteLine ("Numbe! " & i)
    next
End Sub
End Module
```

4. 8 controls & properties

1. Button - Triggers an action when clicked
   - Determines if button can be clicked - Enabled
   - Buttons position - Location

2. Label - Displays static text or info
   - Text - sets text to display
   - Fore color - sets text color

3. Textbox - Allows user to input or edit text
   - Multiline - Multiple lines of text
   - Read only - Prevents editing

4. Checkbox - Provide a yes/no or true false selection
   checked - Indicates its selected
   Checkstate Allows three state, unchecked, checked indeterminate

5. Radiobutton - Allows selection from a group of options
   - Groupname - Groups radio buttons
   - Checked Indicates if selected

6. Combobox - Provide a drop-down menu/list for selection
* items - collection of items
* selected Index - Index of selected item
* Dropdown Style - set the style

7. Listbox - Displays a list of items for selection
* items
* selected Index - Index of selected item
* Multiselect - multiple items selection
* SelectionMode - one, one range selection behavior

8. Picture Box - Displays images
* Borderstyle - sets border type
* Visible - visibility or no visibility